Department of Electrical Engineering
University of Arkansas

# ELEG 3143 Probability & Stochastic Process
# Ch. 2 Discrete Random Variables

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**Dr. Jingxian Wu**
**wuj@uark.edu**

# OUTLINE

- **Random Variable**
- **Discrete Random Variable**
- **Families of Discrete Random Variables**
- **Moments**

# RANDOM VARIABLE

- **Random variable (RV)**
  - A function, which assigns a unique numerical value to each possible outcome in the sample space of a random experiment
  - A mapping from the sample space to some real numbers.

sample space S ➔ real number R

# RANDOM VARIABLE

- **Random variable**
  - Examples
    - Toss a coin. S = {H, T}
    - Define random variable X such that
      - X = 1, if the outcome is H
      - X = 0, if the outcome is T
    - A mapping from the sample space to some real numbers
      - Event {H} → X = 1
      - Event {T} → X = 0
    - The value of the random variable, X, is random. The actual value of X depends on the outcome of the experiment
    - We can assign probability to each value of the random variable
      - Pr(X = 1) = Pr({H}) = 0.5
      - Pr(X = 0) = Pr({T}) = 0.5

# RANDOM VARIABLE

- **Example**
  - Consider an experiment of tossing 2 fair coins. Define the RV, Y, as the number of heads appearing.
    - Find the mapping from the sample space to the value of Y
    - Find the probability of the different values of Y

- Capital letter is used to represent an RV
- Small letter is used to represent a deterministic value of an RV.

# RANDOM VARIABLE

- **Example**
  - Suppose that there is an unfair coin with a probability $p$ of coming up heads. We perform an experiment, in which we flip the coin, until the first head appears. Define an RV, $N$, as the number of flips required.
    - 1. Find the mapping from the sample space to the RV
    - 2. Find the probability of the RV.

# RANDOM VARIABLE

- **Discrete RV v.s. continuous RV**
  - Consider a random variable X
  - If X has countable number values, then X is a discrete RV
    - E.g. toss a coin, X = 0 if H, X = 1 if T
    - E.g. Defined the RV *N* as the number of flips required to get the head of a coin
  - If X has uncountable number of values, then X is a continuous RV
    - E.g. Define the RV X as the current temperature
- **Example**
  - Consider the lifetime of a car
    - Define the RV, X, as the lifetime of the car
      - Then X is a continuous RV, $X \in [0,+\infty)$
    - Define the RV, Y, as follows
      - If the car has a life longer than 2 years → $Y = 1$
      - If the car has a life shorter than 2 years → $Y = 0$
      - Then $Y$ is a discrete RV, $Y \in \{0,1\}$

# OUTLINE

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# DISCRETE RV

- **Discrete RV**
  - A random variable that can take at most a countable number of possible values.
- **Probability mass function (PMF)**
  - Consider a discrete RV *X*, which can take any one of the following values: $a_1, a_2, \cdots, a_n, \cdots$
  - The probability mass function of *X* is defined as

$$p_X(a_i) = \Pr(X = a_i) \qquad i = 1, 2, 3, \cdots$$

  - Example
    - Flip an unfair coin with the probability *p* of coming up head. Define an RV X, such that H → X = 1; T → X = 0. What is the PMF?

# DISCRETE RV

- **Example**
  - Consider an urn with 3 black balls, 5 red balls, and 8 white balls. Randomly pick 1 ball. Define an RV X as:
    - black ball → X = 1, red ball → X = 2, white ball → X = 3
    - Find the PMF.

# DISCRETE RV

- **PMF**
  - For an RV $X \in \{a_1, a_2, \cdots, a_n, \cdots\}$, since X must take one of the values of $a_n$

$$\sum_{i=1}^{\infty} p_X(a_i) = \sum_{i=1}^{\infty} \Pr(X = a_i) = 1$$

# DISCRETE RV

- **Cumulative distribution function (CDF)**
  - All called distribution function

$$F_X(x) = \Pr(X \le x)$$

  - The probability that the RV, *X*, is less than or equal to a deterministic number, *x*
    - capital letter X: Random variable (has random values)
    - Small letter x: deterministic variable (has fixed value)
    - E.g.

$$F_X(3) = \Pr(X \le 3)$$

# DISCRETE RV

- **Example**
  - Flip an unfair coin with the probability *p* of coming up head. Define an RV X, such that H → X = 1; T → X = 0. What is the CDF?

# DISCRETE RV

- **Example**
  - Suppose X has a PMF given by

$$p_X(1) = \frac{1}{2} \qquad p_X(2) = \frac{1}{3} \qquad p_X(3) = \frac{1}{6}$$

    - Find and plot the CDF.

# DISCRETE RV

- **Properties of CDF**
  - $0 \le F_X(x) \le 1$
  - $F_X(-\infty) = 0$
  - $F_X(\infty) = 1$
  - $F_X(x)$ is a non-decreasing function of $x$
  - $\Pr(x_1 < X \le x_2) = F_X(x_2) - F_X(x_1)$
  - $\Pr(X > x) = 1 - F_X(x)$

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# DISCRETE RV

- **Bernoulli RV**
  - Consider an experiment, whose outcome can be classified as either a "success" or as a "failure". The probability of success is *p*. Define an RV X such that
    - Success → X = 1
    - Failure → X = 0.

    Then X is called a Bernoulli RV
  - The PMF

$$\Pr(X = 1) = p \qquad \Pr(X = 0) = 1 - p$$

  - The CDF

  - Recall Bernoulli trials
    - Perform the experiment independently for *N* times

# DISCRETE RV

- **Binomial RV**
  - Perform *n* independent trials, each of which results in a "success" with probability *p*. Define the RV Y as the total number of successes that occur in the *n* trials. Then Y is a binomial RV.
  - PMF

$$p_Y(k) = \Pr(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

  - The summation of PMF

$$\sum_{k=0}^{n} p_Y(k) =$$

# DISCRETE RV

- **Relationship between Bernoulli RV and Binomial RV**
  - Consider an experiment, whose outcome can be classified as either a "success" or as a "failure". The probability of success is *p*. The Bernoulli RV, X, is defined as
    - Success → X = 1
    - Failure → X = 0.

$$X \sim \text{Bernoulli}(p)$$

  - Perform *n* independent trials, each of which results in a "success" with probability *p*. Define the RV Y as the total number of successes that occur in the *n* trials. Then Y is a binomial RV. $X_i$
    - Each trial results in a Bernoulli RV

$$X_i \sim \text{Bernoulli}(p) \qquad Y \sim \text{Binomial}(n, p)$$

  - The relationship between the Bernoulli RV $X_i$, and the Binomial RV $Y$

$$Y = \sum_{i=1}^{n} X_i$$

# DISCRETE RV

- **Example**
  - It is known that any item produced by a certain machine will be defective with probability 0.1, independently of any other item. What is the probability that in a sample of 3 items, at most one will be defective?

# DISCRETE RV

- **Binomial RV**
  - On a multiple-choice exam with 4 possible answers for each of the 5 questions. What is the probability that a student would get 4 or more correct answers just by guessing?

# DISCRETE RV

- **Geometric RV**
  - Suppose that independent trials, each having probability $p$ of being success. The experiment is performed until a success occurs. Define $X$ as the number of trials required until the first success. Then X is said to be a geometric RV.
  - PMF

$$p_X(k) = \Pr(X = k) = (1-p)^{k-1} p$$

  - The sum of Geometric PMF

$$\sum_{k=1}^{\infty} \Pr(X = k) =$$

# DISCRETE RV

- **Poisson RV**
  - A random variable X, taking on the values 0, 1, 2, ……, is said to be a Poisson random variable with parameter $\lambda > 0$, if the PMF can be written as

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

- **Sum of Poisson PMF**

$$\sum_{k=0}^{+\infty} e^{-\lambda} \frac{\lambda^k}{k!} =$$

$$e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

# DISCRETE RV

- **Example**
  - If the number of accidents occurring on a highway each day is a Poisson RV with parameter $\lambda = 0.1$. What is the probability that at least 1 accident occur today?

# DISCRETE RV: SUMMARY

- **Bernoulli RV**

$$P(X=1)=p \qquad P(X=0)=1-p$$

- **Binomial RV**

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$$

- **Geometric RV**

$$P(X=k)=(1-p)^{k-1}p$$

- **Poisson RV**

$$P(X=k)=e^{-\lambda}\frac{\lambda^k}{k!}$$

# OUTLINE

- Random Variable
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# MOMENTS

- **The expected value, or mean, of a discrete RV, *X*, is**

$$E(X) = \sum_{x_i} x_i \, p_X(x_i)$$

  - The summation is performed over all the possible values of *X*.
  - Also called as: expected value, mean, average, first moment
  - Interpretation
    - The *weighted average* of the random variable
    - If the random variable takes all the values with equal probability, that is, $p_X(x_1) = \cdots = p_X(x_N) = \frac{1}{N}$, then

$$E(X) = \frac{1}{N} \sum_{i=1}^{N} x_i$$

    - Given more weight to more probable values, and vice versa
  - Also denoted as

$$E(X) = \overline{X} = m_X$$

# MOMENTS

- **Example**
  - Consider an unfair coin with Pr(H) = 0.3. Define the RV X, such that X = 0 if H, and X = 1 if T. Find the mean of X. (Bernoulli RV)

  - Find the mean of a geometric RV with parameter *p*.

# MOMENTS

- **Example**
  - Find the mean of a binomial RV with parameters *n*, *k*, and *p*

# MOMENTS

- **Example**
  - Find the mean of the Poisson RV with the parameter $\lambda$

# MOMENTS

- **Expected value of any function of X**

$$E[g(X)] = \sum_{x_i} g(x_i) p_X(x_i)$$

  - $E(X)$: the expectation operator
  - The variable of the expectation operator must be a random variable
    - X is an RV
    - g(X) is an RV
  - Also denoted as $E[g(X)] = \overline{g(X)}$

# MOMENTS

- **Example**
  - Consider a gambling game of throwing 3 fair coins. Let X be the RV denoting the number of heads. The payoff is $X^2$, e.g. the Casino pays \$0 for 0 H, \$1 for 1H, \$4 for 2H, and \$9 for 3H. If the Casino charges \$3.5 per play, will you play the game?

# MOMENTS

- **The expectation operator is a linear operator**
  - Consider two random variables X and Y and two constants a and b, we have

$$E[aX + bY] = aE[X] + bE[Y]$$

    - Proof:

# MOMENTS

- **Variance**
  - The variance of a discrete RV X is

$$\mathrm{Var}(X) = \sigma_X^2 = E\left[(X - m_X)^2\right] = \sum_{x_i} (x_i - m_X)^2 \, p_X(x_i)$$

    - $m_X = \mathbb{E}[X]$

  - Also called: the 2nd central moment of *X*
  - Standard deviation:

$$\sigma_X = \sqrt{\mathrm{Var}(X)} = \sqrt{\sum_{x_i} (x_i - m_X)^2 p_X(x_i)}$$

  - Physical interpretation
    - How far away the random variable is from its expected value
    - A larger variance ➔ more uncertainty
    - Zero-variance ➔ the RV is deterministic (no randomness)

# MOMENTS

- **Example**
  - Consider a discrete RV with PMF

$$p_X(x) = \begin{cases} 1/4, & x = 0, \\ 1/2, & x = 1, \\ 1/4, & x = 2, \\ 0, & \text{otherwise} \end{cases}$$

    - Find the mean and variance.

# MOMENTS

- **Properties of variance**
  - $\sigma_X^2 = E[X^2] - m_X^2$
  - $\mathrm{Var}[aX + b] = a^2 \mathrm{Var}[X]$

# MOMENTS

- **Example**
  - Find the variance of the Bernoulli RV
  - Consider a deterministic variable x = 3. What is the mean and variance?

# MOMENTS

- **The *n*-th moment of the random variable X**

$$E[X^n] = \sum_{x_i} x_i^n \, p_X(x_i)$$

  - 1st moment: $E[X]$ mean
  - 2nd moment: $E[X^2]$

- **The n-th central moment of the random variable X**

$$E[(X - m_X)^n] = \sum_{x_i} (x_i - m_X)^n \, p_X(x_i)$$

  - 1st central moment: $E[X - m_X] =$
  - 2nd central moment: $E\left[(X - m_X)^2\right] = \sigma_X^2$ variance

- **The combination of all the moments, n = 1, 2, …., gives a complete description of the random variable**